

國立陽明交通大學 113 學年度碩士班考試入學招生試題

科目：線性代數與離散數學(8102)

考試日期：113 年 2 月 2 日 第 2 節

系所班別：資訊聯招

第 1 頁, 共 2 頁

【不可使用計算機】\*作答前請先核對試題、答案卷(試卷)與准考證之所組別與考科是否相符!!

First you need to write down your answer clearly and then explain how to compute the answers. You also need to answer the questions in order. Do not jump around. *5009*

1. (5 points) Solve the following puzzle.

一名醫護人員說：「醫院裡的醫護人員，包括我在內，總共是16名醫生與護士。並且下面四條狀況都成立。並且如果把我排除不計的話，下面四條狀況仍然都成立。」

- 一、護士多於醫生。
- 二、男醫生多於男護士。
- 三、男護士多於女護士。
- 四、至少有一位女醫生。

請問說話的人是甚麼性別和職務？

請注意，每個人要麼是男性，要麼是女性，不可以又是男又是女。每個人要麼是醫生，要麼是護士，不可以又是醫生又是護士。

2.

a. (5 points) Consider the following non-decreasing sequence of natural numbers  $a_1, a_2, a_3, \dots$ : 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 5, 5, 5,  $\dots$ . Note that there are exactly  $k^2$  occurrence of  $k$ . Define a function  $f(n)$  as the largest integer  $m$  such that  $a_m = n$ . For example,  $f(3) = 14$  and  $f(4) = 30$ . What is  $f(f(n))$ ?

b. (5 points) Find all non-negative integers  $a$  and  $b$  so that  $\sqrt{a} + \sqrt{b} = \sqrt{2009}$ .

3.

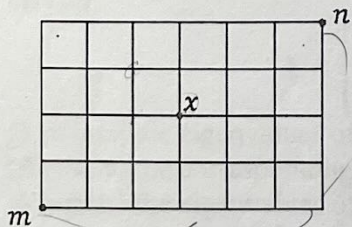
a. (5 points) The *Goldbach's conjecture* says, “every even integer greater than 2 is a sum of two prime numbers.” Please translate Goldbach's conjecture into a logical formula using  $\forall$ ,  $\exists$ , and  $(\wedge)$ , or  $(\vee)$ , not  $(\neg)$ , implies  $(\rightarrow)$  and if-and-only-if  $(\leftrightarrow)$ , and the four basic integer arithmetic operators  $(+, -, *, /)$ . Note that  $/$  is integer division, for example,  $5/2 = 2$ .

b. (5 points) A *bit string* is a string of 0 and 1 digits. For instance, 0011101110 is a bit string of length 10. How many bit strings of length 10 contain either five consecutive 1's or five consecutive 0's?

4. **(11 points)** We want to count the number of paths from vertices  $m$  to  $n$  in the following graph. Each path is made up of a series of steps, where each step is a move one unit to the right or a move one unit upward (No moves to the left or downward are allowed.)

a. (3 points) What is the number of paths from vertices  $m$  to  $n$ ?

b. (4 points) What is the number of paths from vertices  $m$  to  $n$  that do not go through vertex  $x$ ?


$$2009 \times (2009 - 4\sqrt{ab}) = (a-b)^2$$

c. (4 points) Let  $P$  be the set of all vertices in the above graph. Define a binary relation  $R$  on  $P$  such that  $(p, q) \in R$  if we need either one unit to the right or one unit upward to move from vertices  $p$  to  $q$ . Is  $(P, R)$  a partially ordered set? Why or why not?



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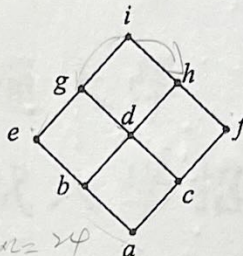
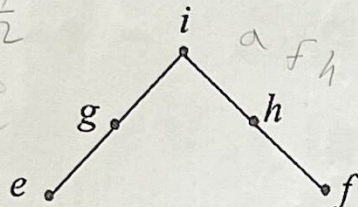
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5. (6 points) If we run topological sorting on the following Hasse diagrams, how many different results can we get?

(2 points) (a)

(4 points) (b)

$$\frac{10}{12} + \frac{-64}{12} = \frac{-54}{12}$$



6. (8 points) Suppose that we already have the following theorem.

If  $G$  is a connected planar simple graph with  $e$  edges and  $v$  vertices, where  $v \geq 3$ , then  $e \leq 3v - 6$ .

Using this theorem to prove that if  $G$  is a connected planar simple graph with at least three vertices, then  $G$  has a vertex of degree not exceeding five.

7. (12 points) Let

$$A = \begin{bmatrix} 0 & -3 & 3 & 9 \\ 2 & 3 & -1 & 1 \\ 1 & 3 & -2 & 2 \\ 4 & 10 & -5 & 1 \end{bmatrix}$$

- a. (6 points) Find the  $A^{-1} = ?$   
 b. (6 points) Find the  $PA=LU$  factorization for  $A$ , where  $P$  is a permutation matrix,  $L$  is a lower triangular matrix with unit diagonal, and  $U$  is an upper triangular matrix.  
 8. (8 points) Given 4 points  $(t, b)$ :  $(0, 20)$ ,  $(1, 0)$ ,  $(2, 0)$ , and  $(3, 0)$ , what is the sum of squared error  $\|e\|^2$  if we fit the closest quadratic polynomial  $b = C + Dt + Et^2$  by the least square approximation?  
 9. (5 points) Prove that if a square matrix  $A$  is invertible, then  $(A^{-1})^T = (A^T)^{-1}$ . (Please leave it blank if you don't know the correct answer, or you will get at most minus 5 points (until questions 7, 8, and 9 are 0 points) for the wrong answer. 不會寫請留白, 答錯最多倒扣 5 分, 扣至 7, 8, 9 題 0 分為止。)  
 10. Given

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2A & 2A \\ A & 3A \end{bmatrix}, \text{ please find}$$

- a. (5 points) the eigenvalues of  $A$ ,  
 b. (5 points) the corresponding eigenvectors of  $A$ , and  
 c. (5 points) the eigenvalues of  $B$ .  
 11. (5 points) Let  $C = \begin{bmatrix} 9 & 3 & -6 \\ 3 & 4 & 1 \\ -6 & 1 & 7 \end{bmatrix} = LL^T$ , where  $L$  is a  $3 \times 3$  lower triangular matrix. Please find the  $L$ .  
 12. (5 points) Let  $D = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & i \\ 0 & -i & 1 \end{bmatrix} = F^H F$ , where  $F$  is a  $3 \times 3$  Hermitian matrix. Please find the  $F$ .

$$3 \ 1 \ -2$$

$$06 \ 02 \ -4$$